

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question
Which quadratic equation has no real solutions?

- Answer**
- A. $x^2 + 14x - 49 = 0$
 - B. $x^2 - 14x + 49 = 0$
 - C. $5x^2 - 14x - 49 = 0$
 - D. $5x^2 - 14x + 49 = 0$

Correct Answer: D

Rationale
Choice D is correct. The number of solutions to a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is greater than zero, then the quadratic equation has two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is less than zero, then the quadratic equation has no real solutions. For the quadratic equation in choice D, $5x^2 - 14x + 49 = 0$, $a = 5$, $b = -14$, and $c = 49$. Substituting 5 for a , -14 for b , and 49 for c in $b^2 - 4ac$ yields $(-14)^2 - 4(5)(49)$, or -784 . Since -784 is less than zero, it follows that the quadratic equation $5x^2 - 14x + 49 = 0$ has no real solutions.

Choice A is incorrect. The value of the discriminant for this quadratic equation is **392**. Since **392** is greater than zero, it follows that this quadratic equation has two real solutions.

Choice B is incorrect. The value of the discriminant for this quadratic equation is **0**. Since zero is equal to zero, it follows that this quadratic equation has exactly one real solution.

Choice C is incorrect. The value of the discriminant for this quadratic equation is **1,176**. Since **1,176** is greater than zero, it follows that this quadratic equation has two real solutions.